

TRACE ELEMENT DISTRIBUTION (Cu, Ga, Ge, Cd, AND Fe) IN SPHALERITE FROM THE TENNESSEE MVT DEPOSITS, USA, BY COMBINED EMPA, LA-ICP-MS, RAMAN SPECTROSCOPY, AND CRYSTALLOGRAPHY

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ABSTRACT

The zinc mines of Tennessee are known to contain high concentrations of certain trace elements (Cu, Ga, Ge, Fe, and Cd). Here, we investigate and compare sphalerites from two mining districts in the center (Central Tennessee Mines: CT) and east (East Tennessee Mines: ET) of Tennessee in order to (1) understand the substitution mechanisms that control the incorporation of trace elements in sphalerite, and (2) relate these mechanisms to the conditions of metal deposition. Electron microprobe and LA-ICP-MS analyses reveal complex chemical zoning in the sphalerites together with coupling of two main groups of trace elements: copper, germanium, and gallium; and iron and cadmium. The zoning is interpreted in terms of substitution mechanisms using different equations, including those previously described, such as $2\text{Cu}^+ + \text{Ge}^{4+} \leftrightarrow 3\text{Zn}^{2+}$, $\text{Cu}^+ + \text{Ga}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, $\text{Fe}^{2+} + \text{Cd}^{2+} \leftrightarrow 2\text{Zn}^{2+}$, as well as new equations involving Ge^{2+} and vacancies: $\text{Ge}^{2+} \leftrightarrow \text{Zn}^{2+}$, $\square + \text{Ge}^{4+} \leftrightarrow 2\text{Zn}^{2+}$, and $\square + 2\text{Ga}^{3+} \leftrightarrow 3\text{Zn}^{2+}$.

Electron microprobe analyses of euhedral crystals of sphalerite highlight a correlation between the presence of trace elements and preferential crystallographic growth directions: copper and germanium are preferentially incorporated along the [010] growth direction, while cadmium is preferentially incorporated along the [110] growth direction. Raman spectroscopy highlights for the first time the possible coexistence of two different crystallographic forms of ZnS in some samples: cubic sphalerite and a non-cubic form, possibly a wurtzite-type structure. This is supported by the LA-ICP-MS data, which demonstrates a relationship between the trace element content and crystallographic structure: the non-cubic form of ZnS is enriched in copper, germanium, and gallium, while iron and cadmium are the most abundant trace elements in the cubic sphalerite. Although most of the main substitution mechanisms are observed in sphalerites from the two mining districts, the sphalerites differ in terms of their germanium grade and valency. These differences can be interpreted both in terms of fluid composition and redox conditions, suggesting that two different episodes of mineralization occurred in Tennessee.

Keywords: sphalerite, wurtzite, germanium, gallium, cadmium, EMPA, Raman spectroscopy, LA-ICP-MS, MVT.

INTRODUCTION

The worldwide demand for germanium (Ge), gallium (Ga), indium (In), and cadmium (Cd) has increased in recent years, notably as a result of the

development of certain sectors of the high-tech industry, such as the manufacture of photovoltaic solar cells. These four elements are extracted from different primary sources. Germanium can be extracted from coal or zinc ore, gallium is an important by-

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product of bauxite and zinc ore, indium can be extracted from tin, copper, and zinc ores, and cadmium is extracted from zinc ore alone.

Cadmium is found in all types of Pb-Zn deposit (at an average concentration of 5000 ppm; Cook *et al.* 2009, Ye *et al.* 2011), whereas germanium, gallium, and indium occur in only specific types of deposits and in variable amounts. Germanium is concentrated in Mississippi Valley Type (MVT) deposits (≈ 100 –300 ppm), in Sedex deposits (50 ppm), and in epithermal deposits (10–100 ppm; Höll *et al.* 2007, Cook *et al.* 2009, Ye *et al.* 2011). Indium is concentrated in Volcanogenic Massive Sulfide (VMS) deposits (100–200 ppm; Ye *et al.* 2011), in Sedex deposits (100–300 ppm; Ye *et al.* 2011), and in epithermal deposits (up to 500 ppm; Cook *et al.* 2009, Ye *et al.* 2011). Gallium is present in MVT (≈ 100 ppm), Sedex (≈ 50 ppm), and VMS (≈ 100 –200 ppm) deposits (Cook *et al.* 2009, Ye *et al.* 2011).

The presence of trace elements such as Cu, Ga, and Ge within sphalerite from central Tennessee has been known since the 1970s (Jolly & Heyl 1968). Zinc concentrates derived from these MVT deposits are characterized by highly variable grades of germanium, gallium, and cadmium. The highest Ge and Ga contents are known to occur in the central Tennessee MVT deposits (250 ppm Ge and 800 ppm Ga), whereas the concentrations of these elements are lower in the eastern deposits (<30 ppm Ge and <200 ppm Ga), despite their very similar geological features.

Since the precursor work of Jolly & Heyl (1968), no study has investigated trace-element distributions in sphalerite from the Tennessee MVT deposits using modern micro-analytical techniques. The present study characterizes trace-element distributions (Ge, Ga, Cd, Fe, and Cu) in sphalerite from the central and eastern Tennessee MVT deposits using a combination of optical observations, Raman spectroscopy, electron microprobe analyses, and LA-ICP-MS data. The study aims to (1) highlight the different substitution mechanisms that lead to the incorporation of trace elements in sphalerite and relate the distribution of these elements to the structure of ZnS, and (2) attempt to relate these mechanisms to the conditions of metal deposition. This information allows for a better understanding of the germanium and gallium metal cycles by determining the concentrations of accompanying metals and the conditions under which these deposits are formed.

GEOLOGICAL SETTING

The zinc mines of Tennessee are located between the Mississippi Valley and the Appalachian Mountains in the USA. Two main mining districts can be

distinguished: (1) The mines of central Tennessee (CT), located in the center of the state and in the Nashville dome within the Cincinnati arch (Fig. 1B). This district encompasses three mines: Cumberland, Elmwood, and Gordonsville. (2) The mines of east Tennessee (ET), located in the eastern part of Tennessee and within the Appalachian basin. The ET district, also known as the Mascot-Jefferson City District, contains three mines: Young, Coy, and Immel. In both districts, the Zn mineralization is hosted within Ordovician rocks belonging to the Knox group (Fig. 1A).

The central Tennessee mining district (CT)

Deposits were first discovered in the CT district in 1963 by the New Jersey Zinc Company (Callahan 1977), and they have since been studied by several authors (Jolly & Heyl 1968, Kyle 1976, Callahan 1977, Fischer 1977, Gaylord & Briskey 1983, Gratz & Misra 1987, Gaylord 1995). The mineralization is hosted between the lower and middle members of the Mascot Formation (Gaylord 1995), which is composed of a light-olive to olive-grey finely crystalline dolostone containing 10% altered and unaltered limestone (Fischer 1977). The underlying Kingsport Formation is composed of a massive brown limestone with some interbedded light olive-grey to brownish grey finely crystalline dolostones (Fischer 1977). The mineralization is hosted within a brecciated system that resulted from the dissolution of limestone beds in the Mascot and the Kingsport Formations (Gaylord 1995). The brecciation processes have been studied in depth by Crawford & Hoagland (1968), Hill *et al.* (1971), and McCormick *et al.* (1971), especially in the mines of the ET district, and several episodes of brecciation are known to have occurred. The late breccia comes after the post-Knox unconformity and affected the underlying formation (Kyle 1976, Gaylord 1995). The mineralization is mainly located in the upper part of the dissolution collapse breccia (Kyle 1976, Gaylord 1995), and only a few Pb-Zn veins pass through the post-Knox unconformity and up to the surface in the Mississippian rocks (Jewell 1947, DeGroot 1973). The mineralization consists of dark reddish-brown and yellowish-brown sphalerites (Fig. 2A) associated with calcite, fluorite, baryte, and some traces of galena, pyrite, and marcasite (Gaylord 1995). Calcite is the main gangue mineral (Gaylord 1995). Using semi-quantitative spectrographic analysis, Jolly & Heyl (1968) identified Fe, Mg, Ti, Mn, Ag, Cd, Cu, Ge, Hg, Pb, and Ga as the main trace elements present in the sphalerites of the CT district. The mineralization is also accompanied by bituminous materials and liquid hydrocarbons (Gaylord 1995). The composition of the

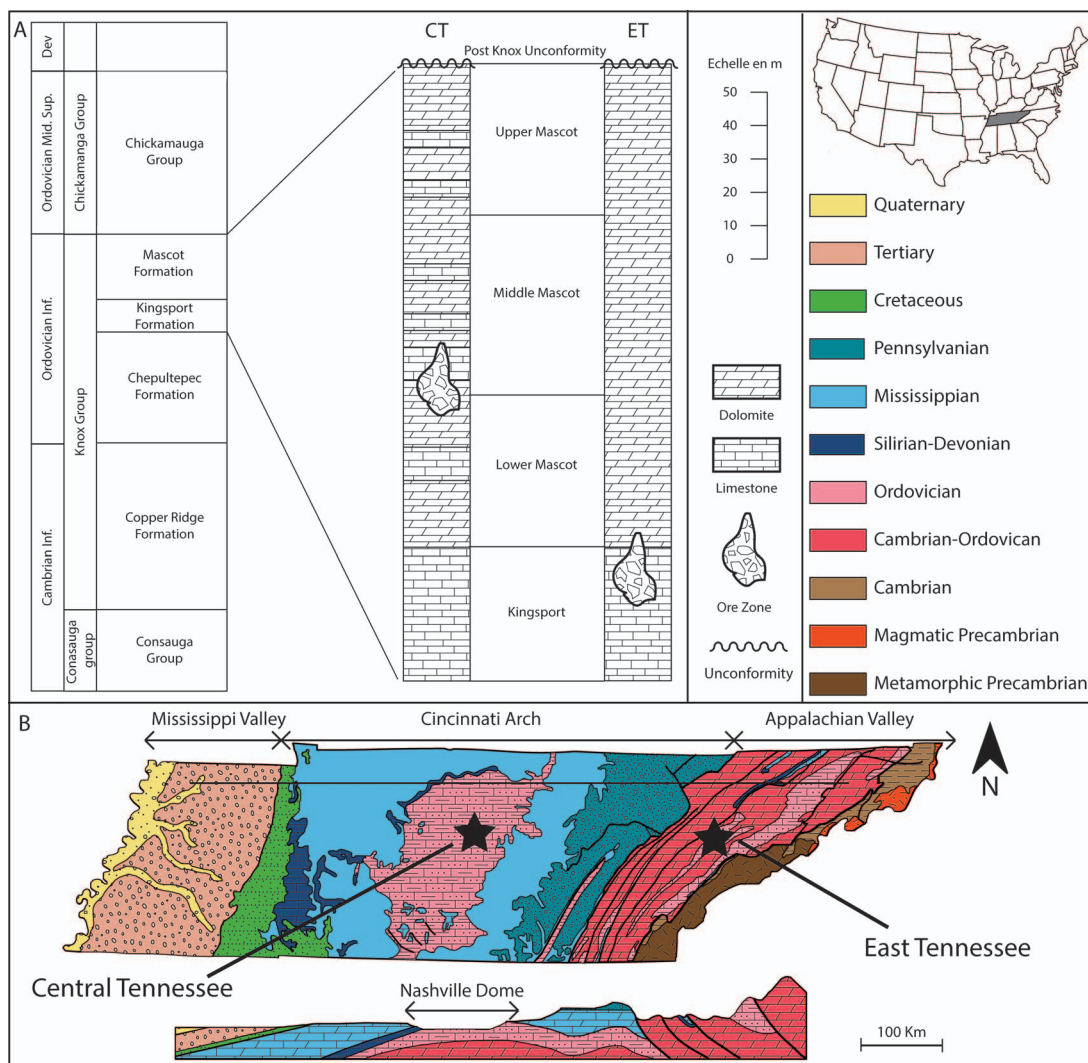


FIG. 1. (A) Stratigraphic column of Tennessee with a focus on the MVT host rocks from central and east Tennessee districts. (B) Geological map and cross-section of Tennessee showing the locations of the two mining districts.

mineralizing fluids has been studied by Gratz & Misra (1987) using fluid inclusions found in the early calcite. The salinity of the fluids ranges from 21 to 23 wt.% NaCl, with homogenization temperatures of between 100 and 130 °C. The mineralizing fluids precipitated both sulfur and carbonate, and therefore were not responsible for the carbonate dissolution (Kyle 1976).

The east Tennessee mining district (ET)

The first mine in the ET district was founded near Jefferson City in 1913 by the American Zinc Lead and Smelter Company (Laurence 1971). The Immel, Coy,

and Young mines were founded later, after World War II (Crawford *et al.* 1969, Hathaway 1969, McCormick *et al.* 1969). The zinc deposits have been studied by several authors (Brokaw & Jones 1946, Crawford *et al.* 1969, McCormick *et al.* 1969). The mineralization is hosted in the upper part of the Kingsport Formation and in the lower part of the Mascot Formation (Brokaw & Jones 1946). In the ET district, the Kingsport Formation is composed of numerous dolostone beds with interbedded limestone in the upper part of the formation, and of massive limestones with interbedded dolostone in the lower part. The deposits are located in an anticline cut by a NE-SW thrust fault

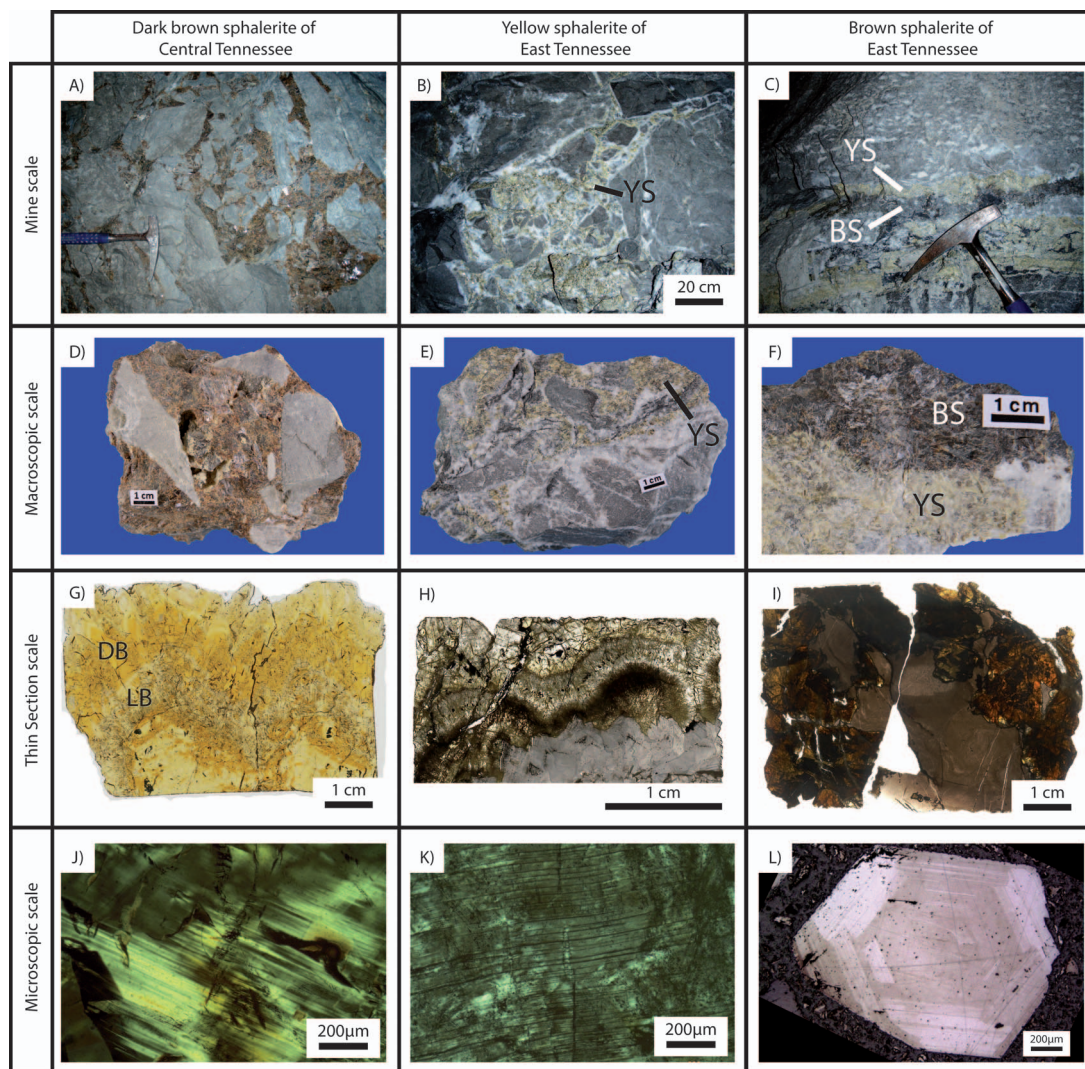


FIG. 2. Samples of sphalerite from Tennessee mines. (A) Brown sphalerite from the CT district filling a brecciated structure. (B) Yellow sphalerite from the ET district in breccia. (C) Brown sphalerite from the ET district in contact with yellow sphalerite (observed only once, at the Coy mine). BS: Brown sphalerite; YS: Yellow sphalerite. (D) Macroscopic sample of brown sphalerite (CT district). (E) Sample of yellow sphalerite (ET district). (F) Sample showing a contact between brown and yellow sphalerites (ET district). (G) Thin section of a brown sphalerite from the CT district exhibiting dark (DB) and light (LB) bands. (H) Thin section of a yellow sphalerite from the ET district showing a "colloform" texture. (I) Thin section of a brown sphalerite from the ET district (only one occurrence). (J) Thin section of sphalerite from the CT district (cross-polarized light) showing the sphalerite anisotropy. (K) Thin section of sphalerite from the ET district (cross-polarized light) showing the anisotropy of sphalerite. (L) Sphalerite crystal observed in reflected light showing zoning after treatment with bleach.

(Brokaw & Jones 1946) that formed during the Alleghenian orogeny (325–260 Ma) (Hatcher & Odom 1980). The mineralization is hosted in a brecciated system that has been described by many authors (Crawford & Hoagland 1968, Hill *et al.* 1971,

McCormick *et al.* 1971). This breccia resulted from collapse of the formation following dissolution of the limestone beds. The dissolution is linked to the post-Knox unconformity in that a karst was formed between the Mascot and Kingsport Formations during immer-

sion of the carbonate platform, which led to dissolution of the limestone beds (Bridge 1956, Harris 1969, Maher 1971). The mineralization mostly consists of sphalerite, with minor galena, pyrite, marcasite and rare fluorite, calcite, and bitumen (Misra & Fulweiler 1995). The majority of the sphalerite from the ET district is Fe-poor (<1% Fe) and pale yellow in color (Fig. 2B). A second type of sphalerite, enriched in iron (>5%) and dark brown in color, has also been described locally (Misra & Fulweiler 1995). The two varieties have been observed together in only one mineralized occurrence in the Coy mine (Fig. 2C). Hill (1969) believed that the dark brown sphalerite represented a later phase of mineralization. The main gangue mineral is dolomite (Misra & Fulweiler 1995). The two main trace elements measured in both of these types of sphalerite are Fe and Cd (Craig *et al.* 1983), and large variations in Fe and Cd contents have been observed. Rb-Sr dating of the sphalerite yielded an age of 377 ± 29 Ma (Nakai *et al.* 1990, 1993), which corresponds to the Acadian orogeny (380–350 Ma). The composition of the fluids has been studied by many authors (Roedder 1971, Zimmerman & Kesler 1981, Taylor *et al.* 1983) through analysis of fluid inclusions in sphalerite. The salinity ranges from 17 to 22 wt.% eq. NaCl and the homogenization temperatures range from 100 to 180 °C. Strontium isotopic studies suggest that the underlying lower Ordovician and lower Cambrian sediments are the sources of the mineralizing fluids (Kesler *et al.* 1988, Furman *et al.* 1990).

Comparison of the east and central Tennessee districts

The two districts share many similarities: (1) zinc mineralization is hosted by the same host rocks: the Mascot and the Kingsport Formations, early Ordovician in age; (2) zinc mineralization fills open spaces in dissolution-collapsed breccia; (3) the collapse is linked to the post-Knox unconformity; (4) the main zinc ore consists of iron-poor sphalerite; (5) sphalerite is associated with minor or traces of dolomite, calcite, galena, pyrite, marcasite, fluorite, baryte, and bitumen; (6) the mineralized fluids have a similar signature with a salinity of around 20 wt.% eq. NaCl and homogenization temperatures between 100 and 150 °C (Roedder 1971, Zimmerman & Kesler 1981, Taylor *et al.* 1983); and finally (7) the lead isotopic signatures are very close (Kesler *et al.* 1994a, Kesler *et al.* 1994b). The main differences between the mines from the two districts concern both (1) tectonic and structural features and (2) mineralogy. With respect to (1), the deposits in the ET district have been disrupted by thrust faulting, whereas those of the CT are relatively flat with no faulting. The CT

deposits also contain many more open spaces and euhedral crystals than the ET deposits. Finally, some/certain sphalerites occur in the Mississippian rocks only in the mines of the CT. With respect to (2), the main gangue mineral is calcite in the deposits of the CT district but dolomite in the ET district. Galena, baryte, fluorite, and organic matter are less abundant in the deposits of the ET district (Gaylord 1995). Finally, the majority of the sphalerite is brown in deposits of the CT districts but yellow in the deposits of the ET district.

SAMPLE DESCRIPTION AND ANALYTICAL METHODS

Sample description

Samples were collected from different mines in the ET and CT districts: Cumberland (CU), Elmwood (ELM), and Gordonsville (GO) in the CT district, and Coy (CY) and Young (YM) in the ET district. Sphalerites were sampled in collapse breccia structures, where they fill open spaces and fractures (Fig. 2A, B). The brown sphalerite from the Coy mine (ET district) was taken from a vein filled with both yellow and brown sphalerite (Fig. 2C).

Macroscopic observations. Sphalerite from the central Tennessee district appears as dark brown well-formed crystals (Fig. 2D) locally associated with euhedral crystals of galena, golden calcite, baryte, and colorless to purple fluorite. Bitumen droplets are frequently observed on the surface of the sphalerite.

In the ET district, the sphalerite is pale yellow in color (Fig. 2E) and occurs as colloform masses. Well-crystallized crystals are rarely observed. Bitumen droplets are also observed but galena is missing. Only one occurrence of brown-dark sphalerite, similar to that described in the mines of the CT district, was observed in the Coy mine, in close association with yellow sphalerite (Fig. 2F). It is not possible to establish which of these two varieties crystallized first.

Microscopic observations. When observed at the microscopic scale with polarized transmitted light, the sphalerite from the CT district exhibits alternating growth zones that range in color from yellow to orange (hereafter referred to as light and dark bands, respectively) (Fig. 2G). Under high magnification, the zoning is seen to be very complex and the crystal edges are hard to define.

The yellow sphalerites from the ET district mainly consist of translucent coarse-grained crystals and, in some samples, the sphalerite appears as a succession of “colloform” bands of 1 to 2 mm thickness, ranging in color from colorless to pale yellow (Fig. 2H). In these samples, the deposition started with dusty sphalerites, rich in tiny fluid inclusions (less than 5 µm), and then

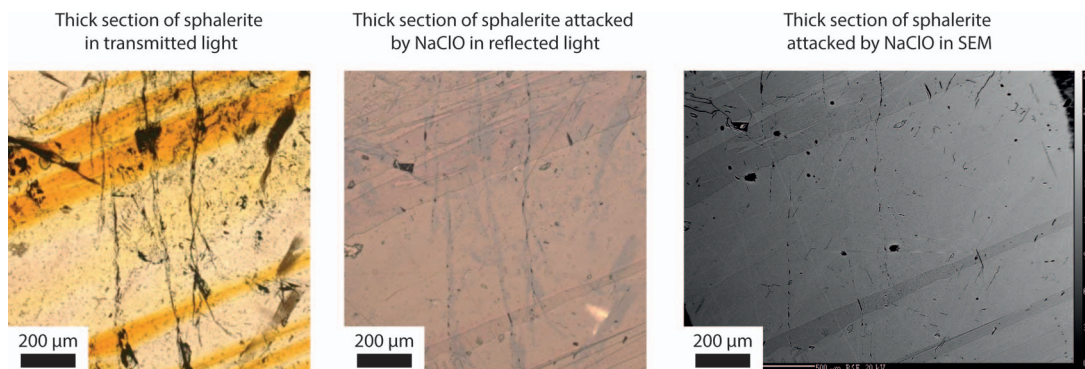


FIG. 3. NaClO etching of a sphalerite from the central Tennessee district.

continued towards more translucent and coarser-grained crystals. In contrast, the brown sphalerites from the ET are homogeneous, with no colored zoning visible in polarized light (Fig. 2I).

In cross-polarized transmitted light, a strong anisotropy can be observed in polished thin and thick sections of the Tennessee sphalerites, ranging from a randomly distributed grid-iron texture to a texture resembling polysynthetic twinning (Fig. 2J, K). These textures, first described in sphalerite from the Elmwood-Gordonsville deposits by Seal *et al.* (1985), are commonly observed in samples from both districts.

The complex zoning in some sphalerites from the CT district can be revealed and improved by etching the surface of the crystal with a dilute hypochlorite (NaClO) solution. The dark bands appear to be preferentially etched (Figs. 2L and 3). In contrast, no zoning was observed in sphalerite crystals from the ET district after etching with NaClO.

ANALYTICAL METHODS

Whole-rock analyses

Whole-rock analyses of concentrates of sphalerites were performed by ALS (Ireland) using two protocols. In the first procedure, ME-ICP-ORE, samples are digested in a mixture of HNO₃, KClO₄, and HBr. The composition of the final solution in dilute aqua regia is then determined by ICP-AES (for Ag, As, Bi, Ca, Cd, Co, Cu, Fe, Hg, Mg, Mn, Mo, Ni, P, S, Sb, Tl, and Zn). In the second procedure, ME-MS61 (48 elements), the samples undergo four near-complete acid digestions and the solution is then analyzed by ICP-MS and ICP-AES (for Ag, Al, As, Ba, Be, Bi, Ca, Cd, Ce, Co, Cr, Cu, Cs, Fe, Ga, Ge, Hf, In, K, La, Li, Mg, Mn, Mo, Na, Nb, Ni, P, Pb, Rb, Re, S, Sb, Sc, Se, Sn, Sr, Ta, Te, Th, Ti, Tl, U, V, W, Y, Zn, Zr).

Electron microprobe analyses

Electron microprobe analyses were performed using a Cameca SX 100 microprobe (GeoRessources laboratory, Nancy - Service Commun de Microscopie Electronique et Microanalyses). The microprobe is equipped with five wavelength-dispersion spectrometers (WDS), four of which contain classic crystal analyzers (TAP, LiF, PET), and one of which is equipped with a large crystal (LLiF and LPET).

Major elements (Zn, S, and Fe) were analyzed with an accelerating voltage of 20 kV, an electron beam current of 20 nA, and a counting time of 10 s for each element (providing a precision of around 0.8% for the measurement). Zn and Fe were analyzed using the $K\alpha$ line and a LiF crystal, and S was analyzed using the $K\alpha$ line and a PET crystal. A Ge-free sphalerite was used (Santander's blende) for calibrating Zn and S contents and a FeS₂ pyrite was used for Fe. Measurements of background noise were made 1500 steps from the peak for S and Zn, and 600 steps from the peak for Fe. Negative values were calculated using a slope of 1.2 for each of the three elements. Germanium was measured using an LLiF crystal following the methodology given in Appendix 1.

The other trace elements were analyzed with an acceleration voltage of 25 kV and an electron beam current of 150 nA. Native copper was used for Cu calibration, Ga was calibrated using GaAs, and cadmium metal was used for Cd. Background noise was measured 600 steps from the peaks for Cu, Cd, and Ga. Negative values were calculated using a slope of 1.2 for Cu and Cd and a slope of 1.1 for Ga. The limit of detection calculated by the software (Cameca - Sight software Peak v5) was about 500 ppm for Fe, 200 ppm for Cu, 180 ppm for Ga, and 400 ppm for Cd.

A total of 349 analyses were obtained from sphalerites from the CT district and 339 from sphalerites from the ET district. Elements such as

Ag, Mn, Sb, Pb, Co, Ni, As, Sn, Te, and Se were also investigated by electron microprobe, but were not detected or were below detection limits with the operating conditions used.

LA-ICP-MS analyses

LA-ICP-MS analyses were obtained at the Georesources laboratory, Nancy, using a 193 nm Microlas Pro Arf Excimer coupled to an Agilent 7500c quadrupole mass spectrometer. The analytical settings for laser ablation and calibration are described in Leisen *et al.* (2012), Lach *et al.* (2013), and Belissont *et al.* (2014). The zinc concentration, determined by electron microprobe analysis, was used as an internal standard. Samples were ablated with a 44- μm spot diameter. The elements measured were ^{57}Fe , ^{63}Cu , ^{71}Ga , ^{74}Ge , and ^{11}Cd with detection limits of 500, 50, 4, 7, and 40 ppm, respectively.

Raman spectroscopy

Raman spectra were recorded using a LabRAM HR spectrometer (Horiba Jobin Yvon) equipped with a 600 g/mm grating and an edge filter (GeoRessources laboratory). The widths of the confocal hole aperture and slit aperture were 500 and 100 μm , respectively. The excitation beam was provided by an Ar⁺ laser (Stabilite 2017, Spectra Physics, Newport Corporation) at 514.53 nm and a power of 100 mW, and was focused on the sample using a 50 $\times$ objective (Olympus). Samples were mapped over areas of 250 $\times$ 320 μm^2 to 950 $\times$ 970 μm^2 using the SWIFT mode ("on-the-fly" mode) in 5 μm steps and with an acquisition time of 0.5 s per step. Raman maps represent the presence or absence of selected peaks, using up to three peaks simultaneously.

RESULTS

Whole-rock analyses

The whole-rock trace-element data for samples and concentrates from mines in both districts are reported in Table 1. All samples are near-pure sphalerite. Iron contents are low and are similar in both districts: between 0.19% and 0.38% in sphalerites from the CT district and between 0.21% and 0.50% in sphalerites from the ET district. One sample from the Coy mine (COY2, Table 1), a brown sphalerite, had an exceptionally high Fe content (1.83%). The cadmium contents of the samples from the two districts are also similar, at around 3000 ppm. With the exception of a concentrate of brown sphalerite from the ET district and a yellow sphalerite from the Young Mine, iron and cadmium show a strong positive correlation in samples from the two districts, with slope close to one (Fig. 4).

Sphalerites from the CT district have higher concentrations of Cu, Ga, and Ge (averages of 782, 798, and 270 ppm, respectively) than those of the ET district (averages of 244, 193, and 13 ppm, respectively). Copper always correlates positively with Ga, with a slope close to one (Fig. 5). The data in Table 1 show that samples from the ET district have higher contents of Mn, Co, and Ni (average of 18.7, 3.7, and 8.4 ppm, respectively) than those from the CT district (averages of 6.6, 1.4, and 0.7 ppm, respectively). The increased Ni and Co in samples from the ET district may be due to the presence of disseminated microcrystals of pyrite, in which these elements were also detected by EDS analysis. The lead content of samples from the CT district is highly variable, but always lower than in samples from the ET district. The lead content may be due to the presence of galena, which is observed at the macroscopic scale in samples from the CT district. Galena was not observed in samples from the ET district, but may be present as inclusions that are too small to be detected with standard microscopic equipment. Except for sample ELM1 (18 ppm Ag), silver is present in very low concentrations in samples from the CT district, and in concentrations of between 1.5 and 4 ppm in samples from the ET district.

Chemical analyses of sphalerite crystals

The Fe, Cd, Cu, Ga, and Ge contents of sphalerites from the ET and CT districts were measured by EMPA, through profiles across bands of different colors (CT) and across successive colloform bands (ET). LA-ICP-MS analyses were also obtained from sphalerites from the CT district, for comparison with the EMPA data, and from sphalerites from the ET district, which are characterized by Ge, Ga, and Cu contents below the EMPA detection limits. Maps of trace element distribution were also made from sphalerites from the CT district.

Determination of trace element contents by electron microprobe. EPMA profiles were made across bands of different colors in sphalerites from the CT district. Compared to the dark bands, the light bands are enriched in Fe and Cd, and characterized by two times lower Fe contents, four times lower Cd contents, and higher concentrations of Cu, Ga, and Ge (Fig. 6). Even though the dark and light bands are easy to distinguish in polarized transmitted light, it is clear that, whatever the thickness of the sphalerite crystals (from 30 to 100 μm), the different color bands are most probably difficult to distinguish on the surface. For example, a dark band covered by a light band will appear dark but will not necessarily outcrop on the surface of the crystal. Analyses of sphalerites from the CT district were therefore obtained from the surfaces

TABLE 1. WHOLE ROCK ANALYSES OF CONCENTRATES OF SPHALERITES FROM DIFFERENT MINES IN TENNESSEE (IN ATOM % AND PPM)

District	Central Tennessee Mine												East Tennessee Mine																						
	Cumberland												Elmwood				Gordonsville				Young				Coy										
	CUM1		CUM1b		CUM2		CUM4		CUM5		CUM6		CUM7		ELM1		ELM3		GORD3		Average		YM1		COY1		COY1b		COY3		COY4		Average		COY2
Sample ID	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	DBS	YS	YS	YS	YS	YS	YS	YS	YS	YS	YS	YS	BS	BS
Type																																			
Zn	%	63.0	2603	64.4	63.6	64.4	66.3	64.1	63.5	64.4	64.7	65.5	64.4	64.4	64.4	64.4	64.4	64.4	64.4	61.5	61.5	60.4	63.2	63.2	63.7	62.0	62.2	62.0	62.2	59.4	59.4				
S	%	31.5	30.3	31.5	32.6	32.6	30.7	30.5	31.4	32.9	32.4	31.6	31.5	31.5	31.5	31.5	31.6	31.5	31.5	30.1	30.1	30.6	30.8	30.8	32.0	30.4	30.8	30.8	30.4	30.8	30.2	30.2			
Fe	ppm	2578	2603	2504	2414	2675	2233	1982	2486	2705	2705	3840	3840	2602	2602	2602	3840	2602	2602	4978	4978	2557	2560	2127	2632	2971	2632	2971	18327	18327					
Cd	ppm	3901	3792	2286	2257	4071	3452	2195	2338	3186	3186	4711	4711	3219	3219	3219	4711	3219	3219	3554	3554	3486	3740	3376	3851	3601	3851	3601	4207	4207					
Cu	ppm	1159	1179	620	586	811	995	1009	213	453	453	798	798	782	782	782	798	782	782	138	138	281	293	195	315	244	315	244	608	608					
Ga	ppm	1268	1288	911	329	824	1030	1210	135	330	330	659	659	798	798	798	659	798	798	149	149	190	194	144	289	193	289	193	23	23					
Ge	ppm	207	245	197	350	209	304	377	298	295	295	218	218	270	270	270	218	270	270	13	13	12	12	12	14	13	14	13	158	158					
Ppb	ppm	126	133	33	45	149	83	47	99	112	112	564	564	139	139	139	564	139	139	197	197	286	284	110	107	197	107	197	362	362					
Co	ppm	1.2	1.2	3.1	1.2	1.0	0.9	1.9	1.9	1.3	1.2	1.1	1.1	1.4	1.4	1.4	1.1	1.4	1.4	4.0	4.0	3.9	4.2	4.2	4.2	2.2	3.7	2.2	3.7	19.1	19.1				
Mn	ppm	<5	<5	<5	<5	<5	<5	<5	6.6	<5	<5	<5	<5	6.6*	6.6	6.6	<5	6.6*	6.6	8.4	8.4	29.2	30.2	8.8	16.9	18.7	18.7	27.8	27.8						
Ni	ppm	0.7	0.6	0.5	0.4	0.5	0.4	0.9	0.9	<0.2	1.7	<0.2	<0.2	0.7*	0.7*	0.7*	<0.2	0.7*	0.7*	9.1	9.1	10.1	10.1	6.5	6.1	8.4	8.4	7.2	7.2						
Ag	ppm	0.7	0.7	<0.5	<0.5	0.9	0.7	3.0	3.0	17.9	5.5	0.9	0.9	3.8*	3.8*	3.8*	0.9	3.8*	3.8*	2.8	2.8	2.1	2.1	2.8	1.5	2.3	1.5	2.3	4.1	4.1					

DBS: Dark Brown sphalerite; YS: Yellow sphalerite; BS: Brown sphalerite.

* Average calculated using a value of 0 for trace element concentrations below the detection limit.

Analyses performed at ALS laboratory.

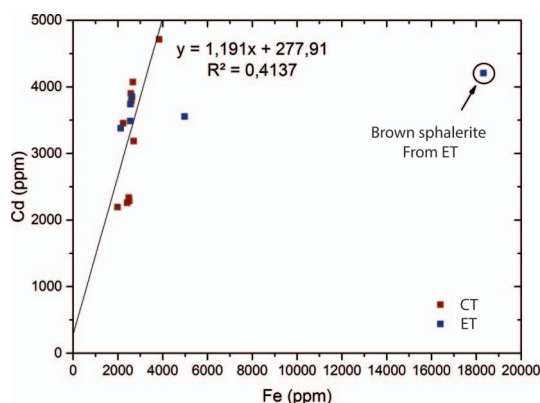


FIG. 4. Cd (ppm) versus Fe (ppm) measured from concentrates of sphalerites from the ET and CT districts (whole-rock analysis).

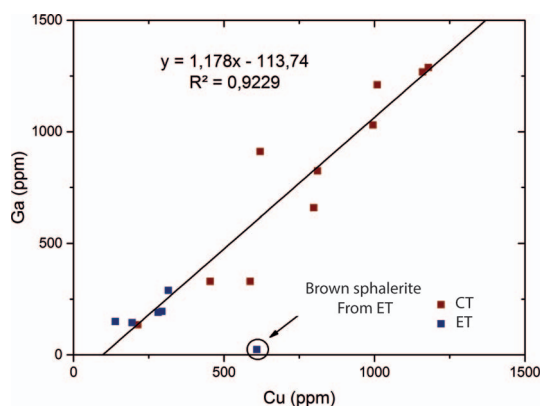


FIG. 5. Ga (ppm) versus Cu (ppm) measured from concentrates of sphalerites from the ET and CT districts (whole-rock analysis).

of crystals that had been treated with NaClO, in order to allow the color band located on the polished surface of the crystals to be revealed when observed with reflected light or by SEM (Fig. 3). In total, 349 analyses were obtained from sphalerites from the CT district (202 analyses of dark bands and 147 of light bands), 339 analyses from yellow sphalerites from the ET district, and 86 from brown sphalerite from the ET district (sample COY 3). Trace element concentrations are summarized in histograms below.

The maps of element distribution in sphalerite from the CT district indicate that the dark bands, as revealed by NaClO etching, are enriched in Ga, Ge, and Cu (Fig. 7B, C, D) whereas the light bands appear to be enriched in Fe and Cd (Fig. 7E, F).

The histograms in Figure 8 (data from EMPA analysis obtained from polished surfaces treated by NaClO) confirm that the light bands in sphalerites from the CT district are relatively enriched in Fe and Cd (concentrations mostly higher than 1500 ppm) compared to the dark bands (<2000 ppm Fe and <1000 ppm Cd). In contrast, the light bands are poorer in Cu and Ge (generally <500 ppm) than the dark bands (up to 1800 ppm Cu and 800 ppm Ge). Gallium contents are comparable in the light and dark bands and are generally below 800 ppm. However, several of the dark bands show an enrichment in Ga of up to 2400 ppm.

The iron contents of yellow sphalerites from the ET district are highly variable (ranging from 0 to 6000 ppm) but always less than those of the brown sphalerites (up to 3.5%). Cadmium concentrations are also variable, between 0 and 3500 ppm in yellow sphalerites and between 1000 and 2500 ppm in brown sphalerites. Copper was detected only a few times in both yellow and brown sphalerites and when present

was generally below 600 ppm. Germanium was rarely detected in yellow sphalerites (two measurements) but was present in brown sphalerites, mostly at around the 200 ppm level. Gallium was not detected.

In order to investigate the correlation between trace element distribution and crystallographic orientation, an additional map of trace element distribution was made from an oriented sphalerite crystal cut perpendicular to the [111] (axis 3) (Fig. 9). Copper was shown to be preferentially incorporated along the [010] growth direction, and cadmium along the [110] growth direction. Copper and Cd are negatively correlated. A higher concentration of germanium is observed at the core of the crystal and is correlated with a high Cu content. Traces of Ga were detected in this crystal but in very low concentrations.

The trace element distribution maps and EMPA spot analyses demonstrate that even though the dark bands are the main host region for Cu, Ga, and Ge, these three trace elements are not always systematically present together.

Determination of trace elements contents by LA-ICP-MS. The concentrations of trace elements determined by LA-ICP-MS are presented in histograms in Figure 10. In samples from the CT district, 21 analyses were obtained from dark bands and 78 from light bands. In sphalerites from the ET district, 27 analyses were obtained from brown sphalerites and 92 from yellow sphalerites.

In the CT district samples, the dark bands contain less Fe and Cd (generally below 2000 ppm) than the light bands (wider ranges of concentration and up to 4000 ppm Fe and 5000 ppm Cd). The dark bands are at least as rich in Cu and Ge as the light bands (which contain up to 600 ppm of these elements) but are often much richer, containing up to 1500 ppm Cu and 800

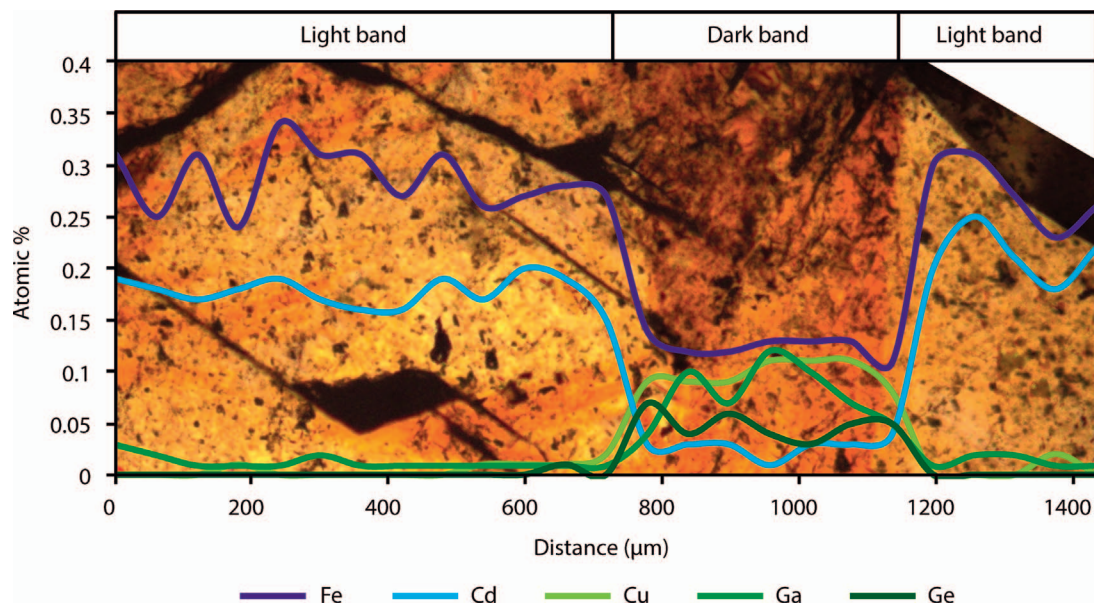


FIG. 6. EMPA profile of the polished surface of a sphalerite from the CT district showing dark and light bands in polarized transmitted light.

ppm Ge. The concentrations of gallium are mostly between 100 and 2000 ppm whatever the color of sphalerite, but up to 4500 ppm was measured in some dark bands.

The yellow and brown sphalerites of the ET district contain very different amounts of iron: less than 2500 ppm in the yellow sphalerites, and more than 5000 ppm in the brown sphalerites. The Cd concentration is also higher in the brown sphalerites (around 3000 ppm) than in the yellow ones (around 1500 ppm). The yellow sphalerites are also poorer in Cu and Ge than the brown sphalerites (yellow: mostly <100 ppm; brown: several hundred ppm), but it should be noted that these elements were detected in very few of the brown sphalerites. Gallium was usually only present in low concentrations (around 100 ppm) in both the yellow and brown sphalerites, but some yellow sphalerite analyses showed higher concentrations (up to 400 ppm Ga).

A comparison of trace-element concentrations measured from the same samples but by two different methods (EMPA and LA-ICP-MS) reveals the same geochemical trends. Although the detection thresholds are lower with LA-ICP-MS, there is very good agreement between the results of the two analytical techniques. In particular, the LA-ICP-MS analyses confirm the very low concentrations of Ga in the yellow sphalerites of the ET district.

Structural analysis of sphalerite crystals by Raman spectroscopy

The anisotropic zones observed in the samples from the CT and ET districts implies the presence of non-cubic zinc sulfide minerals inside the crystals of sphalerite. The only other zinc sulfides currently known to exist are wurtzite polytypes. In order to determine the nature of this anisotropic structure, we used Raman spectroscopy, a method which allows polymorphs to be distinguished. Two types of Raman spectrum, corresponding to the dark and light zones of the sphalerites, characterize the samples from the CT district (Fig. 11). In the spectra for the dark bands, the main peak widens toward lower wave-numbers and there is at least one additional peak between 274 and 285 cm^{-1} . If we compare the spectra to those of different reference materials (R040136 and R100145 for the sphalerite, R060181 for the wurtzite, in the RRUFF database), the light bands are seen to be composed of sphalerite. However, the dark band spectra represent neither sphalerite nor wurtzite of the type presented in the data base and the literature (Brafman & Mitra 1968, Schneider & Kirby 1972). If the sphalerite spectra are subtracted from those of the dark bands, the final spectrum still does not match any of the spectra found in the Raman data bases. Several wurtzite polytypes exist in nature (e.g., 2H, 4H, and 8H), but very few wurtzite references are available in the literature. The dark bands might therefore

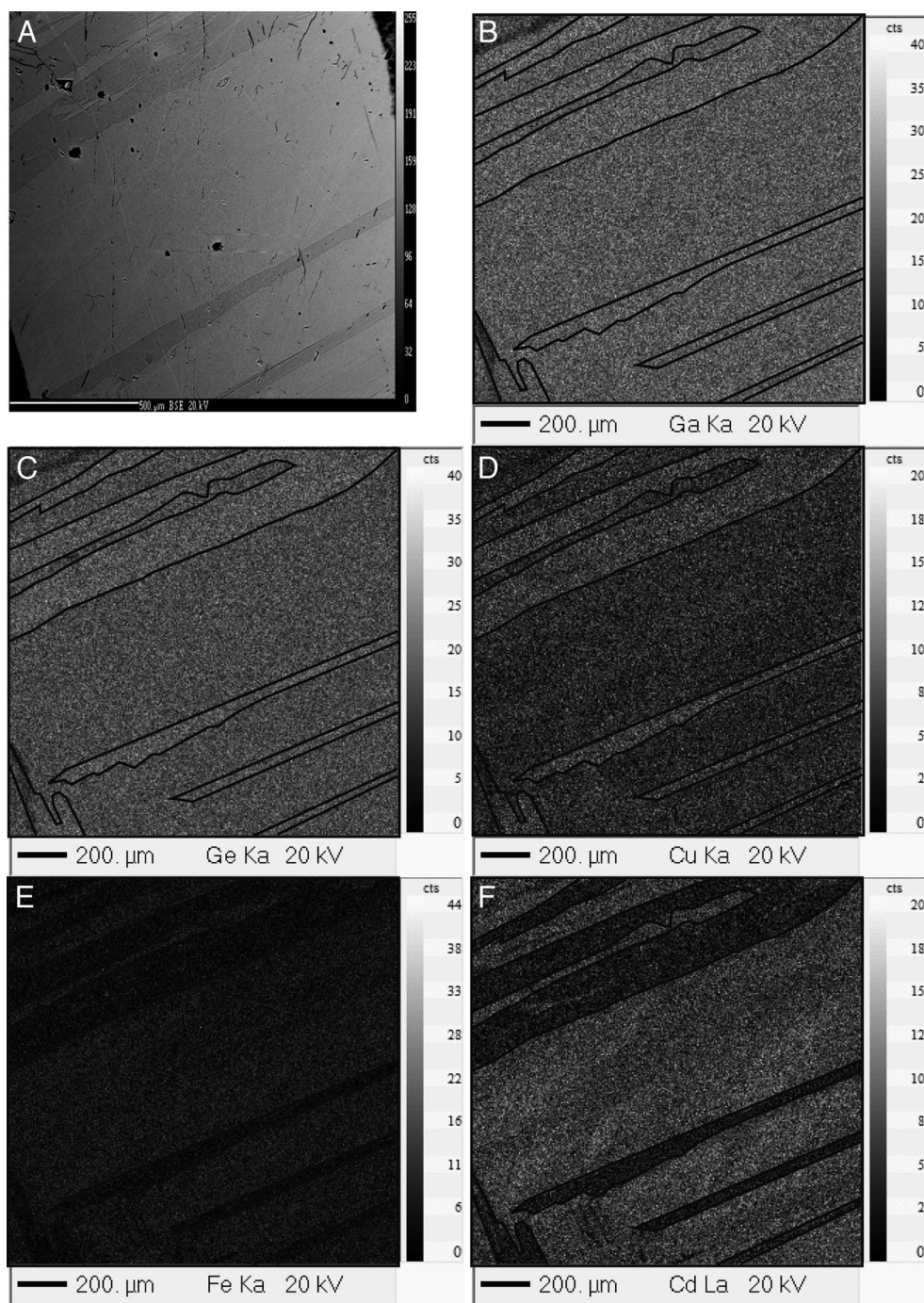


FIG. 7. EMPA maps of trace-element distribution obtained from a polished section of a sphalerite from the CT district: (A) crystal surface observed by optical microscopy in reflected light mode; (B–F) SEM images showing the distribution of (B) Ga, (C) Ge, (D) Cu, (E) Fe, and (F) Cd.

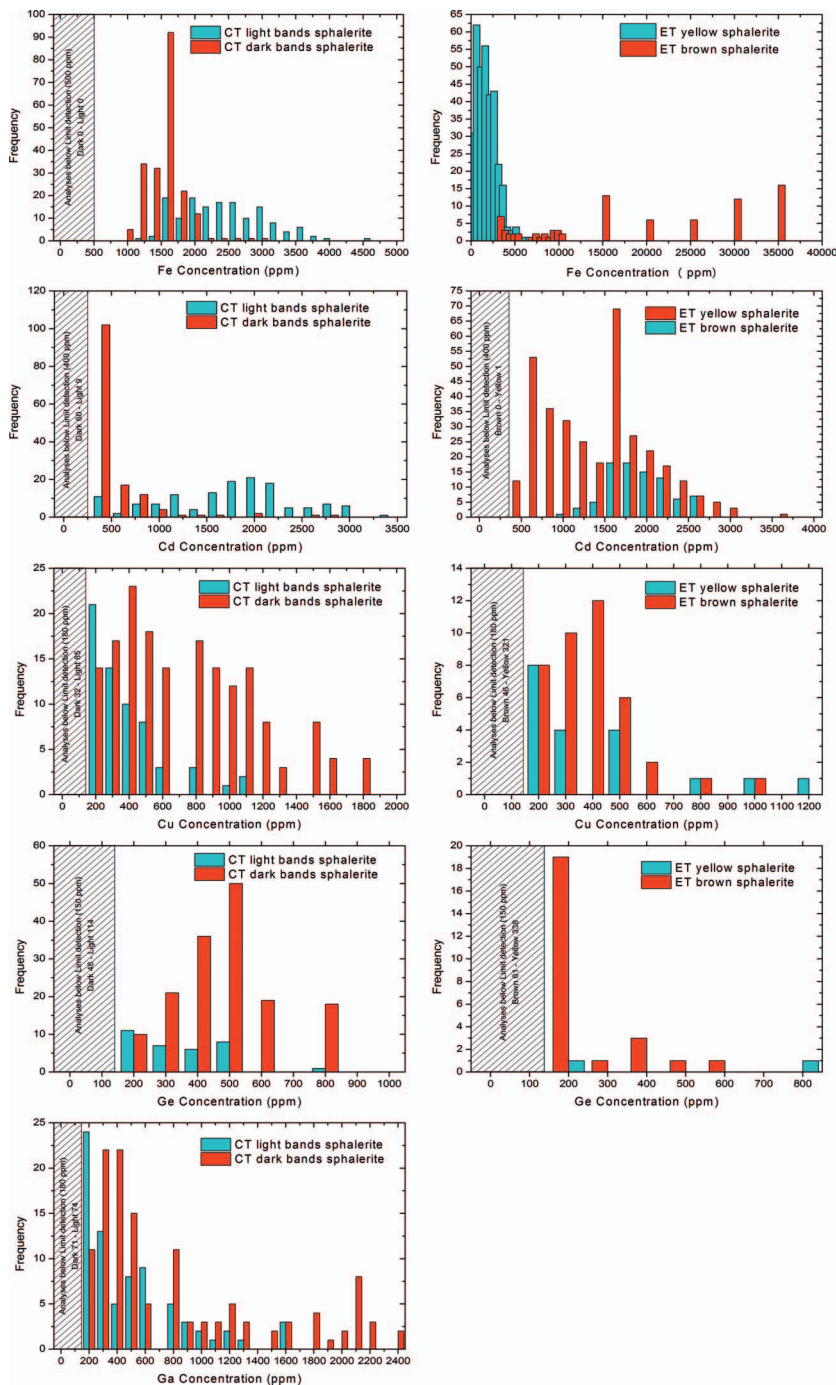


FIG. 8. Histograms showing the distribution of Fe, Cu, Ga, Ge, and Cd in sphalerites from the ET and CT districts. Concentrations determined by electron microprobe analysis.

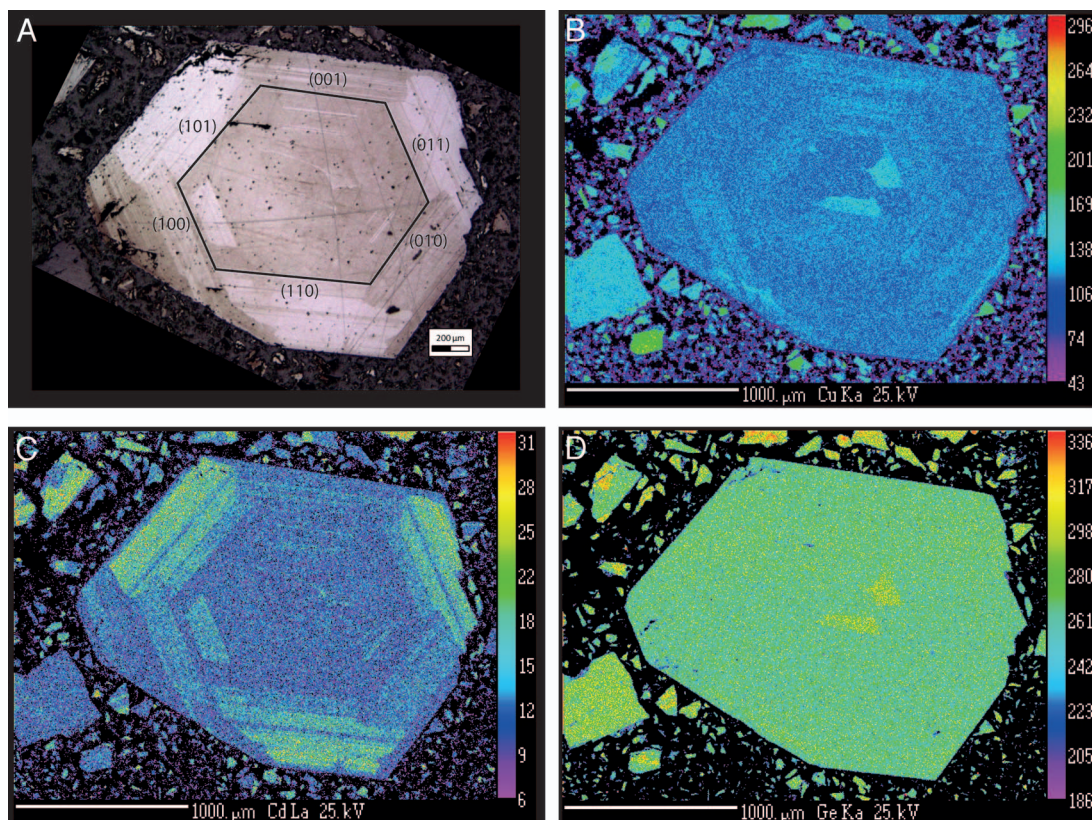


FIG. 9. Electron microprobe mapping of trace-element distribution in a single euhedral crystal of sphalerite from the CT district, cut perpendicular to [111] (a 3-fold rotational symmetry axis). (A) Euhedral crystal observed by optical microscopy in reflected light; the brown zones result from etching with bleach. (B) Map of copper distribution. (C) Map of cadmium distribution. (D) Map of germanium distribution.

correspond to a previously unidentified polytype of wurtzite. The spectra of the dark bands can thus be interpreted either as from a sphalerite which has been deformed by the incorporation of trace elements or that contains stacking defaults, or as a polytype of wurtzite. In the next part of this paper, the spectrum corresponding to the dark bands will be referred to as “wurtzite-type”.

Raman spectroscopic maps were also constructed by integrating the signal obtained over a specific wavenumber range for each of the two Raman signatures: between 345 and 360 cm^{-1} for the sphalerite zones and between 322 and 347 cm^{-1} for the wurtzite-type zones (Fig. 12). For the oriented euhedral crystal of sphalerite from the CT district, the zone enriched in Ge and Cu observed at the core of the crystal exhibits a Raman spectrum similar to that found in the dark bands and characteristic of a wurtzite-type structure. In contrast, the surrounding

parts of the crystal present maps that are typical of the sphalerite structure (Fig. 12). This method thus highlights a correlation between the trace-element compositions of the sphalerites and their crystallographic structures.

An additional transect by Raman spectroscopy was performed on a thin section that exhibits alternating dark and light bands. The transect was then examined alongside a geochemical profile of the section obtained by LA-ICP-MS. The overlapping results confirm the relationship between trace-element composition and crystallographic structure in zinc sulfides (Fig. 13). The cubic structure of sphalerite is dominant when the Ge and Cu contents are lowest and an enrichment in Fe and Cd is observed. In contrast, the wurtzite-type structure appears when Fe and Cd are accompanied by high Cu and Ge contents. Gallium concentrations vary too much to allow it to be correlated with the structure of zinc sulfides.

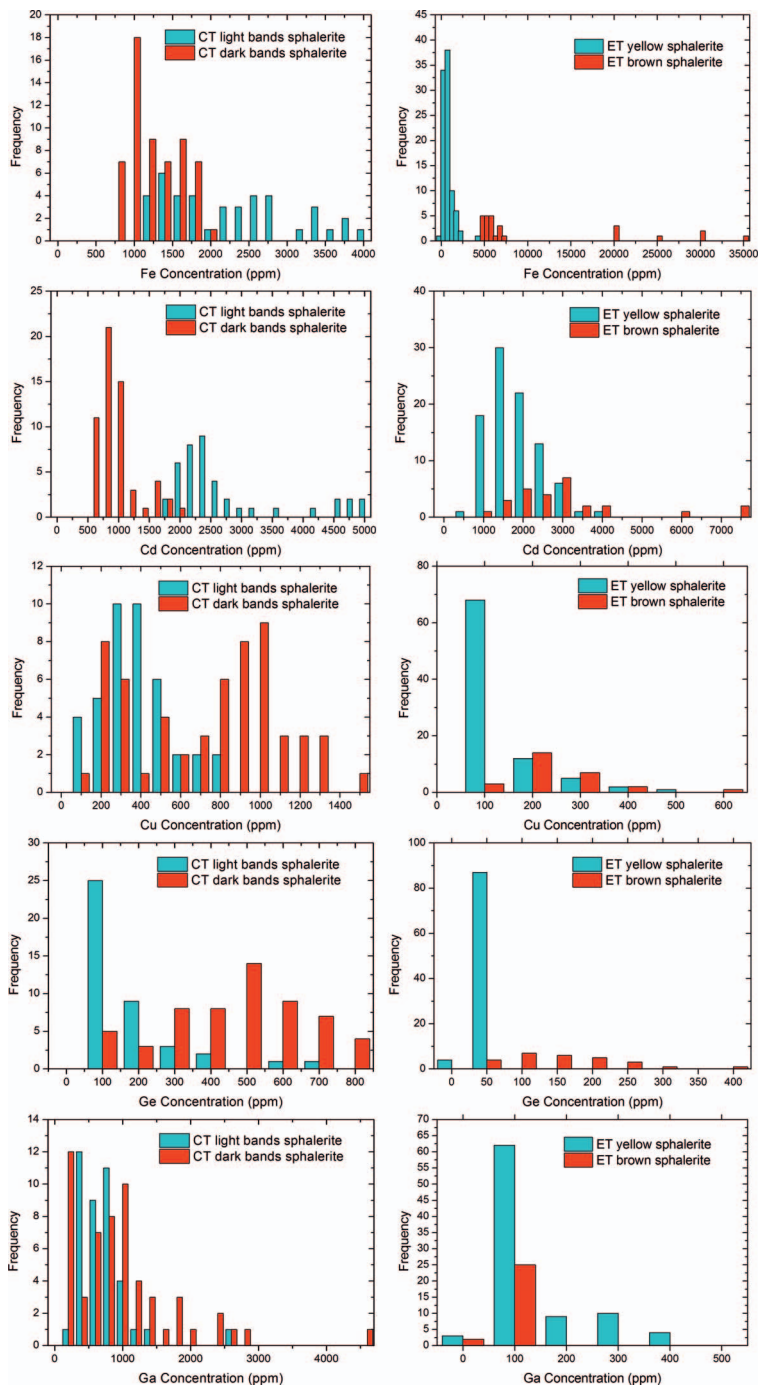


FIG. 10. Histograms of the distribution of Fe, Cu, Ga, Ge, and Cd in sphalerites from the ET and CT districts based on LA-ICP-MS analyses.

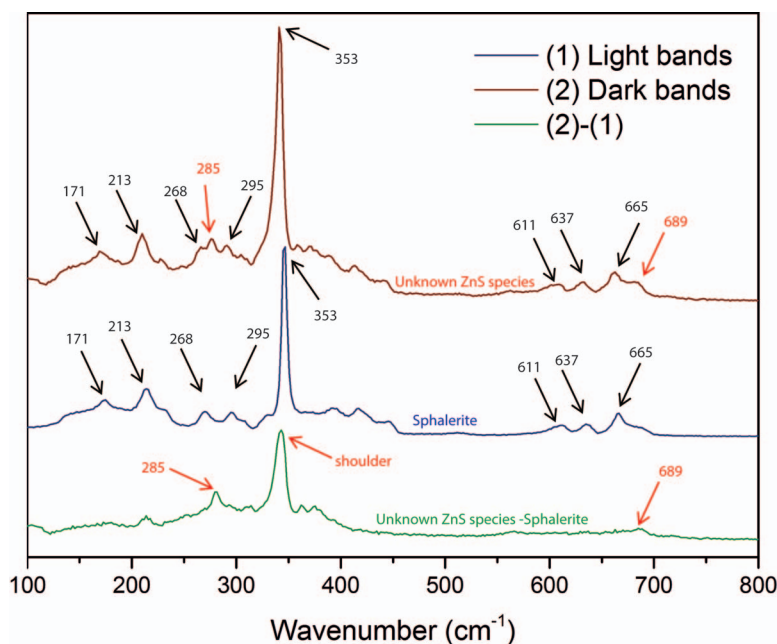


FIG. 11. Raman spectra of samples of the CT district. Blue spectrum: Raman spectrum of light bands and typical of cubic sphalerite. Red spectrum: Raman spectrum of dark bands assigned to an unknown ZnS type. Green spectrum: obtained by subtracting the spectrum of a cubic sphalerite from that of the dark bands.

Raman spectra were also acquired from yellow and brown sphalerites from the ET district. The Raman spectra of the brown sphalerites are characteristic of the wurtzite-type structure while the spectra produced by the yellow sphalerites are typical of a sphalerite structure, similar to those produced by the light bands in samples from the CT district.

DISCUSSION

While numerous authors have studied the incorporation of trace elements in sphalerite [e.g., Beaudoin (2000) on the incorporation of Ag, Sb, and Cu; Benedetto *et al.* (2005) on Mn, Fe, and Cd; and more recently, Pfaff *et al.* (2011) on Ag, As, Cd, Fe, Tl, Sb, Cu, and Mn] few studies have attempted to correlate the color or crystallographic structure of sphalerite with the trace-element chemistry in a way that would enable equations to explain the substitution mechanisms to be proposed.

Trace element signatures in the two mining districts (ET and CT)

The two mining districts studied here are very similar in terms of geology and mineralogy. However, the results of whole rock analyses and electron microprobe and LA-ICP-MS investigations highlight

differences in the distributions of trace elements within the sphalerites. Data obtained by LA-ICP-MS, in agreement with electron microprobe analyses, are summarized in Figure 14a (Fe + Cd versus Cu + Ga + Ge), which highlights the different compositions and different distributions of trace elements in the two types of colored bands in sphalerites from the CT district. In particular, a depletion in Fe + Cd and a potential enrichment in Cu + Ga + Ge are observed in the dark bands.

The Fe + Cd contents of yellow sphalerites from the ET district are lower than those of the brown sphalerites (Fig. 14b) but fall within the same range seen in both dark and light bands in sphalerites from the CT district.

Sphalerites from the CT district have higher concentrations of Cu + Ga + Ge than those of the ET district (Fig. 14a, b), whose Cu contents are broadly similar whatever the color of the sphalerite. Of note, the yellow sphalerites of the ET district are mostly enriched in Ga while the brown sphalerites are enriched in Ge (Fig. 14c). Thus, Ge or Cu seem to be the main elements that control the color of sphalerites of the CT and ET districts.

The whole-rock analyses also reveal several ppm to several tens of ppm of Co, Ni, Pb, and Mn in concentrates of sphalerites from the ET district, while,

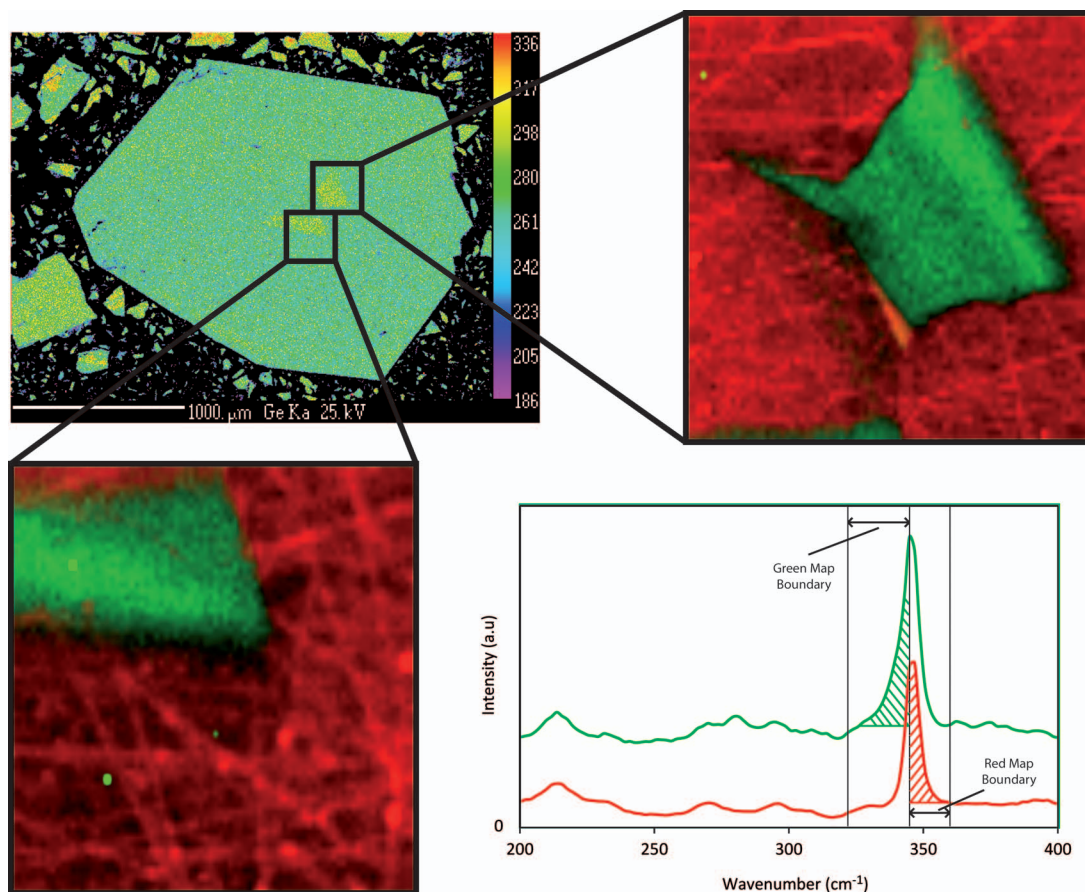
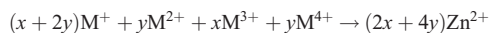


FIG. 12. Raman mapping distribution of sphalerite and wurtzite-type structures in an oriented euhedral crystal of sphalerite from the CT district. Sphalerite zones are in red and wurtzite-type zones are in green.

except for Pb, these elements are present at only very low concentrations in the CT-district sphalerites. These elements were not detected using any of the other analytical methods applied to isolated crystals of sphalerites, thus their presence can be interpreted in terms of the occurrence of microcrystals of galena and pyrite associated with sphalerite in concentrates.

Substitution of zinc by trace elements

Sphalerite incorporates many metals into its structure – Cu^{2+} , Ni^{2+} , Co^{2+} , Fe^{2+} , Mn^{2+} , Ge^{2+} , and Cd^{2+} – all of which have a Z/R ratio of close to Zn^{2+} . These elements substitute for Zn via the simple substitution equation, $\text{M}^{2+} \leftrightarrow \text{Zn}^{2+}$. Johan (1988) proposed a general equation to explain substitution mechanisms for monovalent, trivalent, and tetravalent elements in sphalerite:



where M^+ is Ag or Cu; M^{2+} is Cu, Fe, Cd, Hg, or Zn; M^{3+} is In, Ga, Fe, or Tl; M^{4+} is Ge, Sn, Mo, or W; x is the atomic proportion of M^{3+} ; and y is the atomic proportion of M^{4+} .

A number of studies have used LA-ICP-MS data to propose different substitution equations for trace elements in sphalerite. For example, Cook *et al.* (2009) suggested $\text{Cd}^{2+} \leftrightarrow \text{Zn}^{2+}$, $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, $\text{Ag}^+ + \text{Sn}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, and $\text{Ag}^+ + \text{Cu}^+ + \text{Sn}^{4+} \leftrightarrow 3\text{Zn}^{2+}$; Ye *et al.* (2011) described similar substitution equations with $2\text{Cu}^+ + \text{Sn}^{4+} \leftrightarrow 3\text{Zn}^{2+}$; and Belissont *et al.* (2014) proposed three different substitution equations: $\text{Cu}^+ + \text{Sb}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, $2\text{Ag}^+ + \text{Ge}^{4+} \leftrightarrow 3\text{Zn}^{2+}$, and $\text{In}^{3+} + \text{Sn}^{3+} + \square \leftrightarrow 3\text{Zn}^{2+}$. However, none of these studies established a link between the incorporation of trace elements and the color or crystallographic structure of sphalerite.

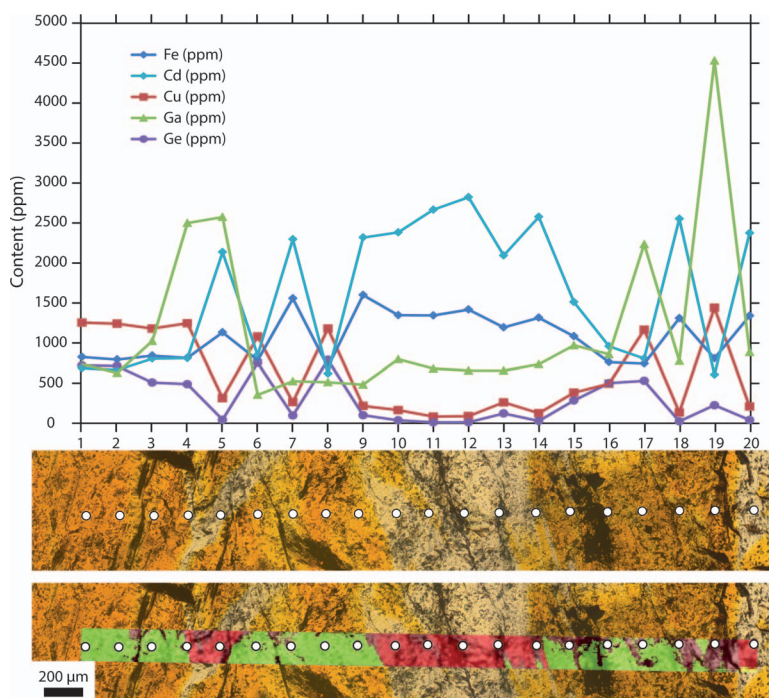


FIG. 13. Raman spectroscopy map showing the distribution of the two structural forms of ZnS (cubic sphalerite in red; wurtzite-type in green) and an LA-ICP-MS geochemical profile of trace elements obtained from the same crystal transect.

Our electron microprobe (spot analyses and mapping) and LA-ICP-MS data demonstrate a relationship between the presence (and concentration) of certain trace elements and the color of the sphalerites. Two groups of trace elements can be distinguished: (1) Fe and Cd, which have no impact on the color of sphalerite if the total concentration of the two elements is less than 0.5 wt.%; and (2) Cu, Ge (and Ga), whose presence leads to a brownish darkening of the sphalerites. In these two groups, correlations were determined between elements that allow the main equations of substitution in the sphalerites of the two districts to be constrained.

Fe-Cd correlations. The yellow sphalerites from the ET district and the dark bands and light bands of sphalerites from the CT district exhibit three main Fe/Cd ratios (Fig. 15). A Fe/Cd ratio equal to 1 is characteristic of the simple substitution equation, $\text{Fe}^{2+} + \text{Cd}^{2+} \leftrightarrow 2\text{Zn}^{2+}$. Johan (1988) pointed out that the incorporation of Cd is facilitated by the replacement of Zn by Fe^{2+} . For sphalerites enriched in Cd ($\text{Fe}/\text{Cd} < 1$) or Fe ($\text{Fe}/\text{Cd} > 1$), two additional simple substitution equations can be proposed: $\text{Cd}^{2+} \leftrightarrow \text{Zn}^{2+}$ and $\text{Fe}^{2+} \leftrightarrow \text{Zn}^{2+}$.

In the iron-rich brown sphalerites from the ET district, the Fe/Cd ratio is always significantly higher

than 1, meaning that the substitution of Zn^{2+} by Fe^{2+} dominates.

Ga-Ge-Cu correlations. For Cu, Ga, and Ge, a number of correlations, and thus substitution equations, can be determined. Following the recommendations of Cook *et al.* (2012), Cu was considered to be monovalent. Germanium was considered both as a divalent or tetravalent cation on the basis of a recent study of germanium speciation in sphalerites from Tennessee by Bonnet *et al.* (submitted). Using X-ray absorption spectroscopy, these authors showed that both forms of germanium, 2+ and 4+, exist in the sphalerites from the CT district, while only the 4+ form is present in the yellow and brown sphalerites from the ET district.

Correlations between Cu, Ga, and Ge are described first in the sphalerites from east Tennessee. Here, the yellow sphalerites are mostly Ga-rich, while the brown sphalerites are generally Ge-rich. These sphalerites are therefore more suitable than those from the CT district for constraining Cu-Ga-Ge correlations and substitution mechanisms.

In Figure 16a, the Ga/Cu ratios of the yellow sphalerites (LA-ICP-MS data) are seen to plot on either side of a positive correlation line with a slope of 1. The following substitution equation is proposed for

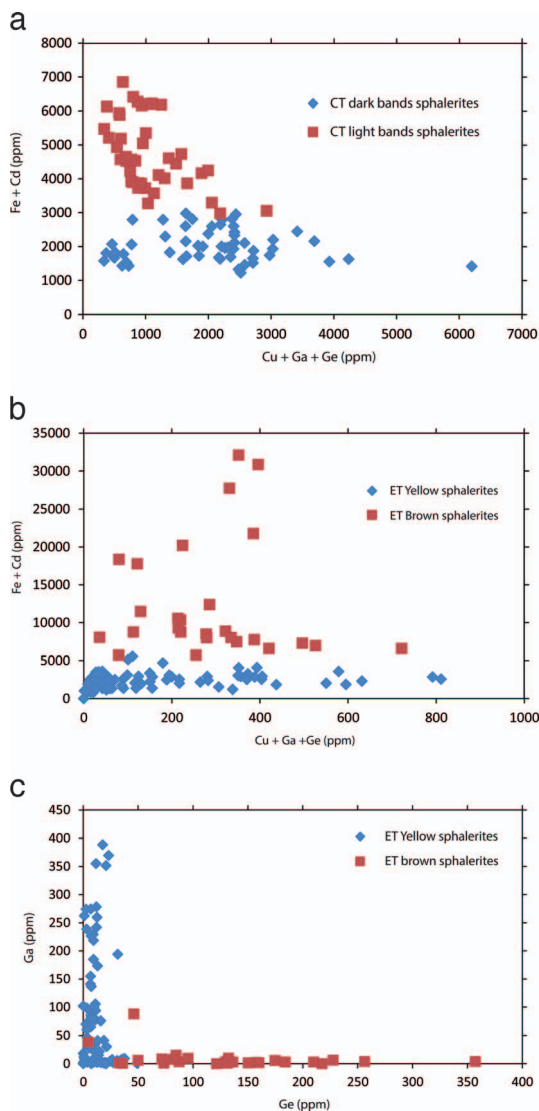
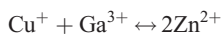
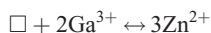


FIG. 14. Trace-element signatures (based on LA-ICP-MS data). (a) Fe + Cd versus Cu + Ga + Ge in sphalerites from the CT district. (b) Fe + Cd versus Cu + Ga + Ge in sphalerites from the ET district. (c) Ga versus Ge in sphalerites from the ET district.

the correlation line:



Additional substitution mechanisms can be suggested for Ga/Cu ratios greater than 1:



and

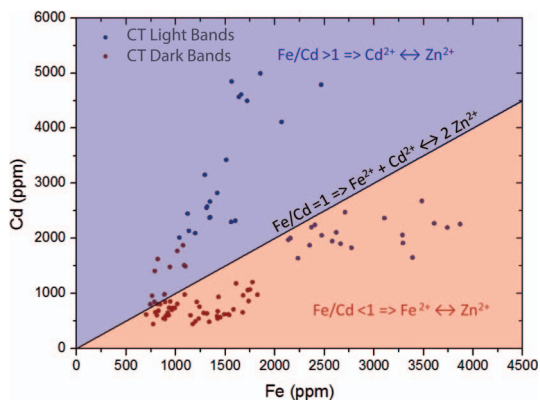
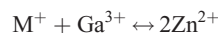


FIG. 15. Correlation between iron and cadmium contents in yellow sphalerites from the CT district and in dark and light bands of sphalerites from the CT district. Data obtained by LA-ICP-MS.



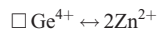
where M⁺ corresponds to metals such as Ag⁺ or Co⁺, which were detected in low amounts in the yellow and brown sphalerite concentrates from the ET district.

Ga/Cu ratios of less than 1 in the yellow sphalerites are due to the existence of an additional correlation between Cu and Ge. In contrast, the brown sphalerites, which are Ga-poor, are characterized by a Ga/Cu ratio of 0.

In Figure 16b, the Ge/Cu ratios of yellow sphalerites are seen to be close to 0 because these sphalerites are Ge-poor. The Ge/Cu ratios of brown sphalerites are always greater than 0.5. A line with a ratio equal to Ge/Cu = 0.5 corresponds to the equation:



For Ge/Cu ratios greater than 0.5, only one additional substitution mechanism is proposed, as only Ge⁴⁺ (and not Ge²⁺) was measured in brown sphalerites by Bonnet *et al.* (submitted):



In the sphalerites from Central Tennessee, both Ga and Ge are present. As Cu is involved in the substitution of Zn together with both Ga and Ge, possible substitution mechanisms can be considered by examining Ga + Ge versus Cu (Fig. 17) and the equations proposed above.

In Figure 17, the (Ga + Ge)/Cu ratios of dark and light bands in sphalerites from the CT district are distributed around a correlation line with a slope of 1. For data that plot below this line, two substitution equations can be proposed:



and

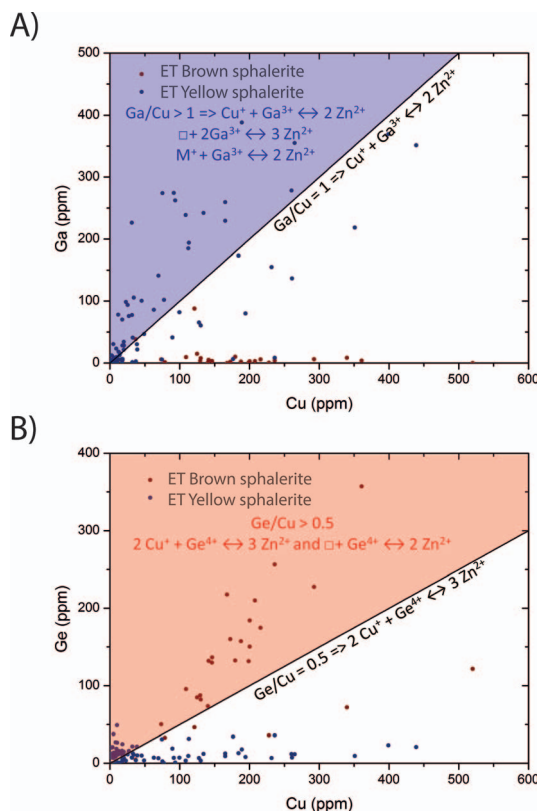
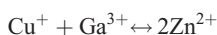


FIG. 16. Correlations between Cu content and (A) Ga content and (B) Ge content, in yellow and brown sphalerites from the ET district.



This line was drawn from the correlations of the trace elements, deduced in the case of the sphalerites of the ET district by combining the two systems, $\text{Cu}^+ + \text{Ga}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ and $2\text{Cu}^+ + \text{Ge}^{4+} \leftrightarrow 3\text{Zn}^{2+}$. These two equations are in good agreement with data reported in other studies (Bernstein 1985, Johan 1988, Cook *et al.* 2009, Ye *et al.* 2011, Cook *et al.* 2012, Belissont *et al.* 2014).

In the $(\text{Ga} + \text{Ge})$ versus Cu diagram, if substitutions only occur with copper, then there are three possible substitution mechanisms. The first corresponds to sphalerite with only germanium and the data are characterized by a line with a slope of 0.5, corresponding to the equation $2\text{Cu}^+ + \text{Ge}^{4+} \leftrightarrow 3\text{Zn}^{2+}$. The second represents sphalerite with only gallium and is characterized by a line with a slope of 1, corresponding to the equation $\text{Cu}^+ + \text{Ga}^{3+} \leftrightarrow 2\text{Zn}^{2+}$. The third possibility represents sphalerite with both germanium

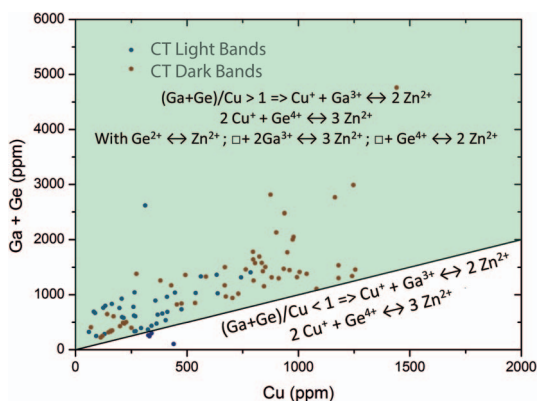


FIG. 17. Correlation between $(\text{Ga} + \text{Ge})$ content and Cu content in dark bands and light bands of sphalerite from the CT district. Data obtained by LA-ICP-MS.

and gallium, and in this case the data plot between the two previous systems, defined by slopes of 0.5 and 1.

When $(\text{Ga} + \text{Ge})/\text{Cu}$ ratios are greater than 1, this is consistent with enrichment only in Ga and Ge. These two elements, sometimes accompanied by structural vacancies, replace Zn. The additional substitution equations are therefore:

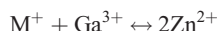
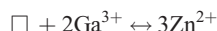
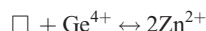
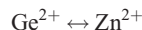


Figure 17 demonstrates that the different substitution mechanisms proposed above may occur together, both in the dark bands and in the light bands of sphalerites from the CT district. However, Zn may be substituted by Cu with Ga and Ge more frequently in the dark bands than it is in the light bands. The substitution of Zn^{2+} by Ge^{2+} could be indicative of reducing conditions and appears to mostly occur in the light bands of sphalerites from the CT district.

Crystallographic controls on trace element incorporation

A relationship between the crystallographic structure of zinc sulfide and the distribution of trace elements has been highlighted. The incorporation and enrichment of metals follows certain specific crystallographic directions in the sphalerite and is also distributed between the cubic sphalerite structure and a wurtzite-type structure.

Copper and Ge are preferentially incorporated along the [010] growth direction of sphalerite while Fe and Cd are incorporated along the [110] growth direction. Our results are in good agreement with those of Johan (1988) but contradict those of Wright (2009), who found that Fe and Cd are preferentially incorporated along [111] growth zones in sphalerite. Because Ga correlates with Cu and Ge, it might also be preferentially incorporated along the [010] growth direction. However, as gallium is trivalent like indium, it might instead be preferentially incorporated along the [111] growth direction, as was reported for indium by Johan (1988). The latter hypothesis would also explain the near absence of Ga in the map we constructed for a section of sphalerite cut perpendicular to a 3-fold rotational symmetry axis (Fig. 9).

For the first time, two types of ZnS structure have been revealed by Raman spectroscopy: the cubic form known as sphalerite, which is the most common and most abundant zinc sulfide mineral in the samples from Tennessee, and a non-cubic form of zinc sulfide, probably a wurtzite-type, that is rarer in these samples. These two different structures explain the strong anisotropy sometimes observed in our samples by optical microscopy. Seal *et al.* (1985) proposed that the anisotropy observed in the sphalerites is caused by a locally high $\text{Cd}^{2+} \leftrightarrow \text{Zn}^{2+}$ substitution rate, leading to clusters of CdS (greenockite), which has a hexagonal structure. This interpretation conflicts with our results, which indicate that the Ge and Cu enrichments are associated with the wurtzite-type structure, and thus the anisotropic structures, whereas the Fe and Cd enrichments characterize the isotropic cubic sphalerites.

Wurtzite is known to be a high temperature form of ZnS, with the sphalerite–wurtzite transition located near 1020 °C at 1 atm (Allen *et al.* 1912). These conditions are incompatible with low-temperature deposits like the MVT. However, it has been suggested that the temperature of inversion between sphalerite and wurtzite might be lowered by the presence of trace elements (*i.e.*, Ga and Fe in Ueno *et al.* 1996). Others have shown that the inversion temperature can also be lower in highly reducing environments (Scott & Barnes 1972). These authors also demonstrated that the temperature of inversion is constrained by sulfur fugacity. A reducing environment and a low sulfur fugacity, combined with the presence of trace elements, in particular Ge and Cu (and Ga), could therefore explain the existence of a wurtzite-type structure in the samples from Tennessee. Wurtzite has been successfully precipitated at room temperature using sulfate-reducing bacteria (Leleu *et al.* 1975), but in general, the BSR (Bacterial

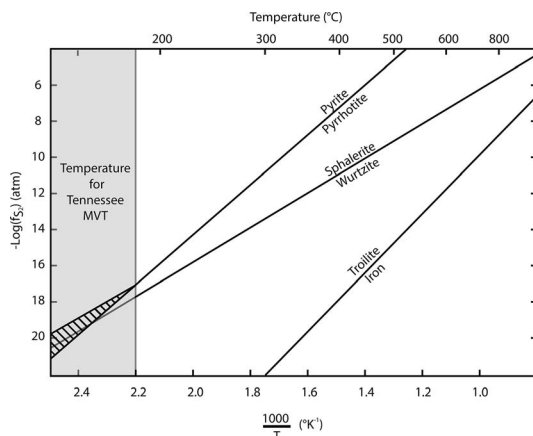


FIG. 18. Temperature ($1000/T$) – sulphur fugacity [$\log(f_{\text{S}_2})$] diagram showing the sphalerite-wurtzite, pyrite-pyrrhotite, and troilite-iron equilibria. The temperature range for the MVT deposits of Tennessee is also shown (adapted from Scott & Barnes 1972).

Sulfate Reduction) is efficient only at temperatures below 100 °C. The existence of a wurtzite-type structure at low temperature, as in the MVT deposits, is thus more likely the result of a combination of the sulfur fugacity and the concentration of trace elements. The nature of this wurtzite-type structure requires further investigation using Raman spectroscopy.

Genetic model for the incorporation of trace elements in sphalerite

In order to explain the geochemical and structural differences between the two types of sphalerite found in the CT district and between sphalerites from the two districts, two hypotheses are proposed. The first is based on a model of two fluids with different trace-element contents and different sulfur fugacities. One fluid precipitates sphalerite and the other precipitates wurtzite. The second hypothesis is based on a model involving a single fluid that precipitates both sphalerite and wurtzite and distributes the trace elements depending on the crystal structure. The complex compositional zoning, which correlates with the alternating sphalerite-wurtzite texture presented in Figure 13, is not consistent with a model involving two fluids. The second hypothesis, involving a single fluid, is thus favored here. This hypothesis was inspired by a number of different studies, for example that of Pfaff *et al.* (2011), which showed that trace elements are not shared equally between the crystalline and colloform forms of ZnS. Beaudoin (2000) reported a similar partitioning of Ag, Sb, Fe, and Cu between

acicular sphalerite, which is inverted from wurtzite precipitation, and massive sphalerite. Patrick *et al.* (1993) also proposed that the distribution of the trace element pairs (Fe + Cd) and (Cu + In), could be due to changes in pH.

As the formational temperature of the ZnS deposits in the mining districts of Tennessee (100–130 °C) has already been determined from fluid inclusion studies (Gratz & Misra 1987), and as sphalerite coexists with both wurtzite and pyrite, the sulfur fugacity can be estimated using the f_{S_2} -temperature diagram of Scott & Barnes (1972) (Fig. 18). The sulfur fugacity of the Tennessee MVT is therefore estimated to be around 10^{-20} atm. This value is close to that determined by Bonnet *et al.* (submitted) from thermodynamic calculations.

CONCLUSIONS

Using a novel approach based on a multi-technique characterization of trace-element distributions in sphalerites, we have obtained important data that significantly improves our understanding of substitution mechanisms and consequently, of the formation of the Tennessee zinc deposits.

Electron microprobe and LA-ICP-MS analyses show that sphalerites from the two different mining districts are characterized by very different copper, germanium, and gallium compositions despite the geological and mineralogical similarities between the two districts. Analyses also show the link between the color and trace-element content. In central Tennessee, more sphalerites appear dark in thin section, as they are rich in Cu, Ga, and Ge, and poorer in Fe and Cd. In east Tennessee, yellow sphalerites are rich in Ga, while brown sphalerites are rich in Fe, Cd, Cu, and Ge.

Even though the same types of substitution mechanisms can explain the incorporation of trace elements in sphalerites from the two districts, a new equation, involving Ge^{2+} , is cited for the first time. The presence of divalent germanium leads to different interpretations in terms of redox conditions of the zinc deposition between the two mining districts.

A strong relationship between the distribution of trace elements and the structure of the ZnS crystals has also been highlighted using electron microprobe and Raman spectroscopy maps. In the sphalerite crystals, copper, germanium, and gallium are preferentially incorporated along the [010] growth direction, while iron and cadmium are preferentially incorporated along the [110] direction. These results, which are not always in agreement with those usually reported in literature, warrant more in-depth analysis. Raman spectroscopy has revealed for the first

time the occurrence of two crystal forms of ZnS inside a single crystal, a cubic form known as sphalerite, and a hexagonal form that could be wurtzite. These two structures are related to specific compositions: the cubic form is richer in iron and cadmium, while the non-cubic form is enriched in copper and germanium (and gallium). Raman spectroscopy coupled with X-ray spectroscopy or LA-ICP-MS appears to be a promising method for studying the distribution of trace elements in sphalerite and understanding the relationship between composition and structure.

This study provided new knowledge of the substitution mechanism in zinc sulfide. First, it has proposed several equations for the substitution of Zn by trace elements (Ga, Ge, Cu, Fe, and Cd). It has revealed a link between the trace-element content and the color of sphalerite, but also with the structure of ZnS. Finally, the presence of several forms of ZnS was used to estimate the sulfur fugacity present during MVT formation.

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